Thm. Let $\{a_n\}$ be a series, and take $R = \left(\limsup_{n\to\infty} |a_n|^{\frac{1}{n}}\right)^{-1}$.

Then, the series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely if $|x| < R$,
diverges if $|x| > R$.

Therefore, we can define $f : (-R, R) \to \mathbb{R} ; x \mapsto \sum_{n=0}^{\infty} a_n x^n$.

For any $0 < \varepsilon < R$, f converges uniformly on $[-R+\varepsilon, R-\varepsilon]$ and f is differentiable on $(-R, R)$, with

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

proof. The first assertion is given by the root test.
And, the second assertion is given by the weierstrass-M test, specifically, for all $n \in \mathbb{N}$ and $x \in [-R+\varepsilon, R-\varepsilon]$, $|a_n x^n| = |a_n| \cdot |x|^n \leq |a_n| \cdot |R-\varepsilon|^n$, and

$$\sum_{n=0}^{\infty} |a_n| \cdot |R-\varepsilon|^n$$

converges, by the first. To show the third assertion, observe that

$$\limsup_{n\to\infty} |a_n|^{\frac{1}{n}} = \limsup_{n\to\infty} |n a_n|^{\frac{1}{n}}$$

Because: since $x_n = n^{\frac{1}{n}} - 1$ satisfies $(1+x_n)^n = n \geq \frac{n(n-1)}{2} \cdot x_n^2$, thus

$$x_n^2 \leq \frac{2}{n-1} \to 0 \quad \text{as} \quad n \to \infty$$

Thus $n^{\frac{1}{n}} \to 1$ as $n \to \infty$. Given $\varepsilon > 0$, there is $N_1 \in \mathbb{N}$ s.t

$$n \geq N_1 \Rightarrow |a_n|^{\frac{1}{n}} < \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} + \frac{\varepsilon}{2}$$

Meanwhile, since $n^{\frac{1}{n}} \to 1$ as $n \to \infty$, take $N_2 \in \mathbb{N}$ s.t

$$n \geq N_2 \Rightarrow n^{\frac{1}{n}} < 1 + r, \quad \text{where } r = \frac{A+\varepsilon}{A+\frac{\varepsilon}{2}} - 1, \quad A = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}}.$$

Then,

$$n \geq \max\{N_1, N_2\} \Rightarrow n^{\frac{1}{n}} \cdot |a_n|^{\frac{1}{n}} \leq \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} + \varepsilon.$$

Moreover, there are infinitely many $n \in \mathbb{N}$ s.t $\limsup_{n\to\infty} |a_n|^{\frac{1}{n}} - \frac{\varepsilon}{2} < |a_n|^{\frac{1}{n}}$,
we can choose similarly, infinitely many $n \in \mathbb{N}$ s.t $1 - t < n^{\frac{1}{n}}$, $t = 1 - \frac{A-\varepsilon}{A-\frac{\varepsilon}{2}}$
Now proved. Using this, $\sum n a_n x^{n-1}$ converges uniformly in $(-R, R)$,
we obtain the result.

Thm. Suppose that $f : (-R, R) \to \mathbb{R} : x \mapsto \sum_{n=0}^{\infty} a_n \cdot x^n$ is well-defined.

Given $c \in (-R, R)$, the equality holds:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} \cdot (x-c)^n \qquad (x \in B_{R-|c|}(c)).$$

Proof. Observe that

$$f(x) = \sum_{n=0}^{\infty} a_n \cdot [(x-c)+c]^n = \sum_{n=0}^{\infty} a_n \cdot \sum_{m=0}^{n} \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m.$$

And, since $x \in B_{R-|c|}(c) \iff |x-c| + |c| < R$,

$$\sum_{n=0}^{\infty} \sum_{m=0}^{n} \left| a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m \right| = \sum_{n=0}^{\infty} |a_n| \cdot [|x-c| + |c|]^n < \infty.$$

So, $\sum_{m=0}^{\infty} |A_{nm}|$ converges for each n, where $A_{nm} = a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m$

we can change the limit,

$$f(x) = \sum_{m=0}^{\infty} \left(\sum_{n=m}^{\infty} \binom{n}{m} a_n \cdot c^{n-m} \right)(x-c)^m = \sum_{m=0}^{\infty} \frac{f^{(m)}(c)}{m!} \cdot (x-c)^m.$$

Abel.

Thm. Let $f: (-1,1) \to \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n x^n$ be defined.

If $\sum_{n=0}^{\infty} a_n$ converges, then $\lim_{x \to 1-} f(x) = \sum_{n=0}^{\infty} a_n$.

Proof. Put $A_n = \sum_{k=0}^{n} a_k$, and $A = \lim_{n \to \infty} A_n = \sum_{n=0}^{\infty} a_n$. As a Cauchy Product,

$$\frac{1}{1-x} f(x) = \sum_{n=0}^{\infty} A_n \cdot x^n, \text{ so } f(x) = (1-x) \cdot \sum_{n=0}^{\infty} A_n x^n \quad (-1<x<1)$$

and since $(1-x) \cdot \sum_{n=0}^{\infty} x_n = 1$, so

$$\begin{aligned} f(x) - A &= (1-x) \sum_{n=0}^{\infty} A_n x^n - A \cdot (1-x) \cdot \sum_{n=0}^{\infty} x_n \\ &= (1-x) \left(\sum_{n=0}^{\infty} A_n x^n - A \cdot \sum_{n=0}^{\infty} x_n \right) \\ &= (1-x) \cdot \left(\sum_{n=0}^{\infty} (A_n - A) \cdot x^n \right). \quad (-1<x<1). \end{aligned}$$

Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t $n \geq N \Rightarrow |A_n - A| < \varepsilon$, and put $M = \max\{|C_n - A| \mid n = 0, \dots, N\}$.

Now, $|f(x) - A| \leq \left| (1-x) \cdot \sum_{n=0}^{N} (A_n - A) x^n \right| + \left| (1-x) \cdot \sum_{n=N}^{\infty} (A_n - A) x^n \right|$

$$\leq (1-x) \cdot MN + \varepsilon$$

Therefore, $\lim_{x \to 1-} f(x) = A$.

이제, 길고도 지루했던 논의는 끝낸다.

Thm. Suppose that $\sum a_n x^n$ and $\sum b_n x^n$ converge in $(-R, R)$.
Let $E \subseteq (-R, R)$ be a set such that

$$x \in E \iff \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} b_n x^n.$$

If E has a limit point in $(-R, R)$, then $a_n = b_n$ for all $n \geq 0$.

Proof. put $c_n = a_n - b_n$, and let $f(x) = \sum_{n=0}^{\infty} c_n x^n \quad (x \in S)$

Then, $f(x) = 0$ on E. Let $A = E' \cap (-R, R)$, and $B = (-R, R) \setminus A$.
Since E' is closed, $B = (-R, R) \cap E'^c$ is open. And, if A is open, then $\{A, B\}$ is separation of an interval $(-R, R)$, so A or B is must be empty.
By the hypothesis, $A \neq \phi$. So, $B = 0$, $A = E' \cap (-R, R) = (-R, R)$.
Now, Since f is continuous in $(-R, R)$ so